XGS Energy

Materials science unlocking geothermal energy globally

Introduction Materials

Q1 2025

Summary

XGS Energy is commercializing cutting-edge proprietary materials science to massively increase the productivity and scalability of geothermal energy.

Our proprietary Thermal Reach Enhancement (TRE) system is 50X more thermally conductive than rock, requires no water and can be deployed *anywhere in the world*.

We will deliver our first commercial projects in late 2025 and are developing a 14 GW greenfield and brownfield project portfolio across 7 countries in North America, Asia, Europe, and the Middle East.

In 2024, XGS completed a 2x oversubscribed $20mm financing round led by Constellation Technology Ventures, Valo Ventures, and VoLo Earth Ventures.

Proven Leadership with Deep Project Experience

XGS Leadership Team

Josh Prueher
President & CEO
B.S., U.S. Naval Academy; M.B.A., U. Oxford

- Founder, CEO & CFO of 3 energy technology & infrastructure companies incl. FlexGen, Broad Reach Power
- Experience deploying over $900mm in infrastructure capital and developing 4+ GW of clean power projects

Ghazal Izadi
Chief Operating Officer
Ph.D. Reservoir Engineering, Penn State

- Global Lead in Reservoir Technical Services at Baker Hughes
- Managed team of 200+, overseeing technical operations for 100+ projects across 20 geothermal fields spanning 5 continents

Axel-Pierre Bois, Ph.D.
Chief Technology Officer
Ph.D. Geomechanics, U. Sherbrooke

- Founder and CEO of CURISTEC technical consultancy for oil & gas and geothermal industries
- Coauthored more than 60 technical papers and holds 10+ patents for novel well completion technologies

Lucy Darago
Chief Commercial Officer
Ph.D. Chemistry, UC Berkeley

- Principal at Boston Consulting Group (BCG), strategic advisor to F500 energy, public sector
- Designed and implemented clean energy procurement strategies for multiple 10+ TWh/year energy consumers

XGS Team Experience

slb · Baker Hughes · HALLIBURTON · bp · NOV · Earl Energy · FlexGen · BCG · Broad Reach Power · CURISTEC · Capuano Engineering Company

World-Class Board and Execution Partners

Board of Directors

Craig Barrett, Ph.D.
Chairman of the Board
Former Chairman & CEO, Intel

Martin S. Craighead
Board Director
Former Chairman & CEO, Baker Hughes; Board Director at Emerson Electric and Texas Instruments

Ray Mabus
Board Director
Former US Sec. of the Navy
Former US Amb. to Saudi Arabia
Former US state Governor

Piotr D. Moncarz, Ph.D.
Founder & Vice Chairman
Adjunct Professor, Stanford
Former Senior Fellow & Principal Engineer at Exponent Inc.

David Seldin
Board Director
Co-founder and Managing Partner, Anzu Partners
35+ years in private equity

Glen Robinson
Board Director
Vice President, Constellation
30+ years in power industry finance, strategy, and operations

Josh Prueher
President and CEO
Founder, CEO and CFO of 3 energy technology companies incl. FlexGen, Broad Reach Power

Khalid Alsweilem
Board Director
Former Chief Invest. Officer, Chief Counselor, and Dir. Gen. of Invest. at Saudi Arabian Monetary Agency (SAMA)

Joe Goodman, Ph.D.
Board Director
Co-founder and Managing Partner, VoLo Earth Ventures

Partners & Investors

Constellation. ANZU PARTNERS VALO ventures VoLo Earth VENTURES THIN LINE CAPITAL NORTHSTAR CLEAN ENERGY HP BASELOAD CAPITAL HYDROGEN SYSTEMS CURISTEC CAPUANO ENGINEERING COMPANY

XGS Formed to Address Challenges to TW-Scale Geothermal

Geothermal, historically, is:

Hard to Find

Existing geothermal tech requires:

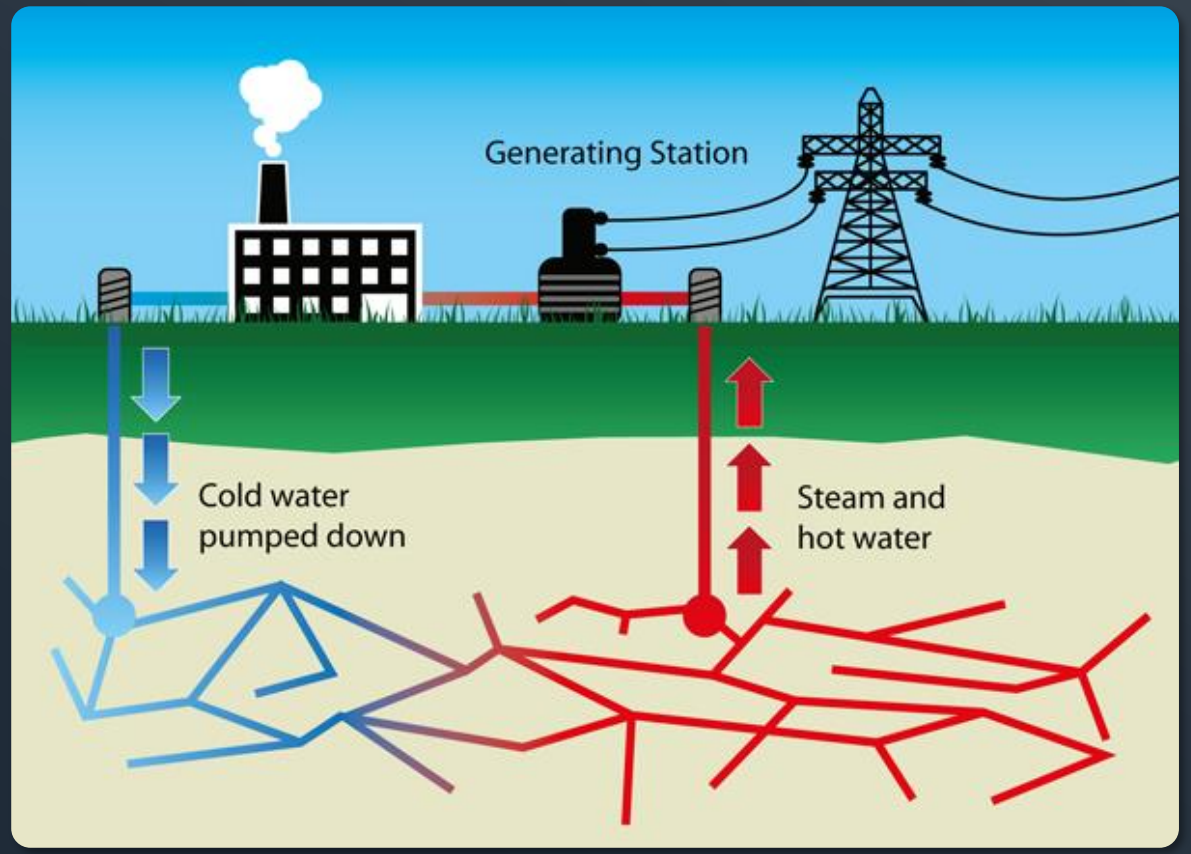
- Hot rock
- Right geology
- Abundant water

Hard to Finance

Operational challenges include:

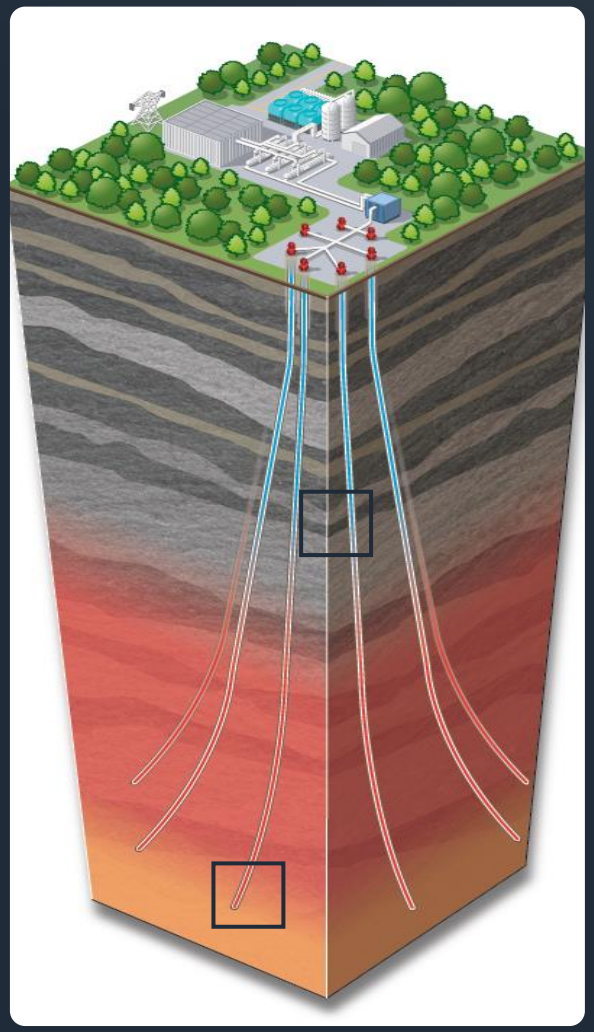
- Subsurface integrity
- Water losses
- Corrosion and scaling

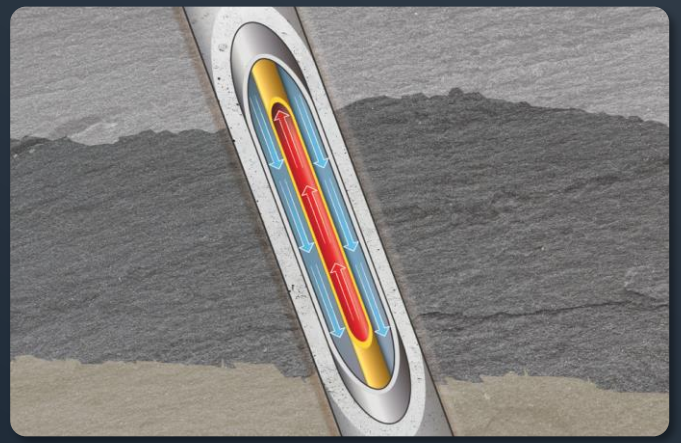
Conventional approach

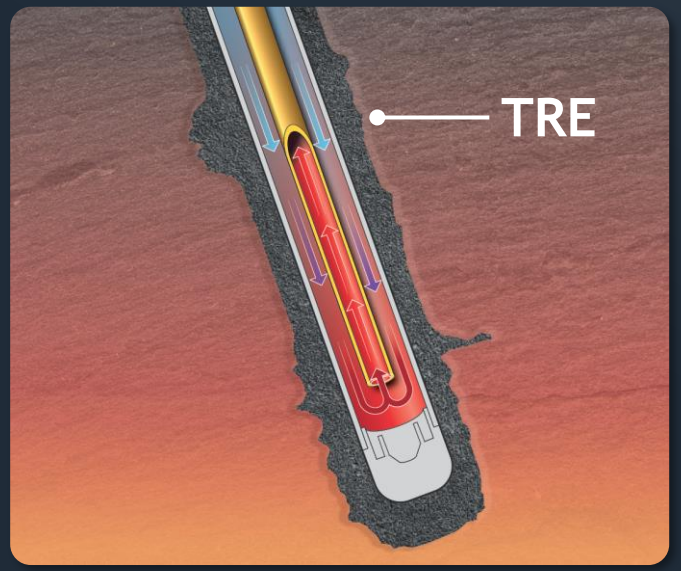


How It Works: XGS Geothermal System

XGS has 70+ awarded and pending patents for its geothermal system







XGS's closed-loop system is both water and geology independent, allowing it to be deployed anywhere in the world

How it works:

- Drill: A single well is drilled to access hot rock using proven oil and gas procedures
- Enhance: Advanced thermally conductive materials are "squeezed" into hot rock around the well
- Close the loop: Hot fluid returned to the surface in insulated inner-tube, creating tube-in-tube closed loop

Proprietary design of Thermal Reach Enhancement (TRE) enables 30-50% more heat capture from hot rock

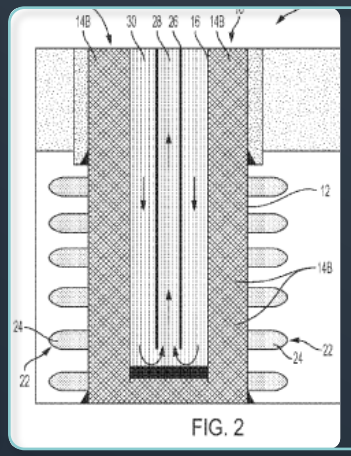
Accelerating Commercial Readiness to Meet Unprecedented Demand

Technology milestones from last 12-months

100-m Shallow Well Demonstration

Validated ability to mix, pump, and place TRE using commercial cementing equipment

Exceeded target TRE heat-transfer properties for commercial readiness, improving by 10% over lab-scale tests



IP Portfolio Progression

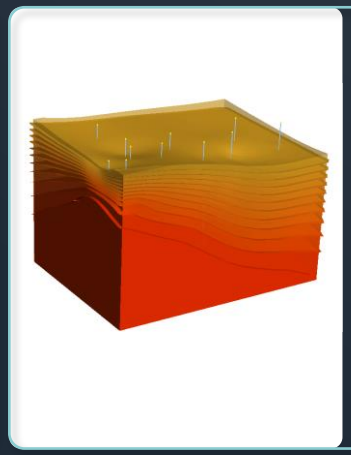
10 patents awarded on TRE concept, compositions, and placement methods in 2024

60+ patents pending for XGS's geothermal production system

Commercial-scale Prototype

Actively deploying full subsurface system including TRE at commercial scale (>1-km, 200C+)

On track for TRE commercial cost target



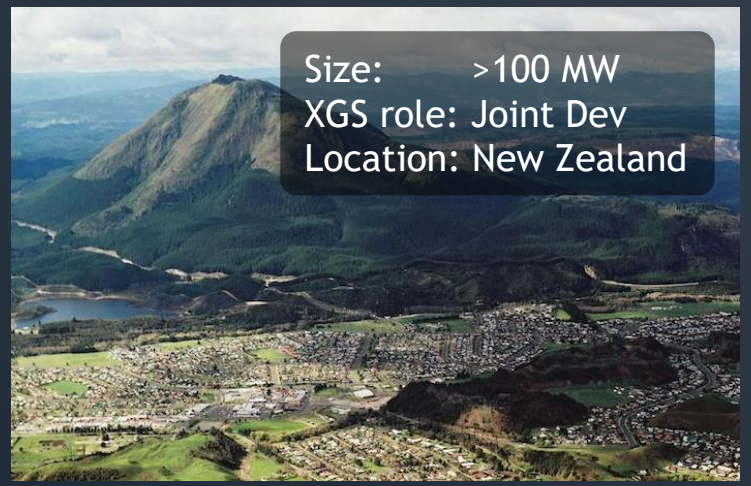
Resource Evaluation

7 priority resource evaluations have identified an incremental ~12GWe of potential

XGS technology is, on average, tripling the energy output potential compared to conventional systems

GW-Scale Project Pipeline Has Significant Offtaker Traction

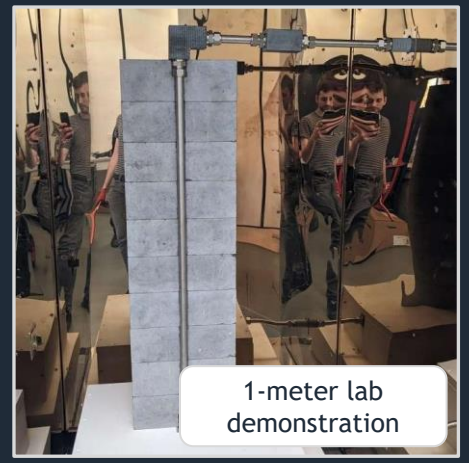
Select projects in pipeline

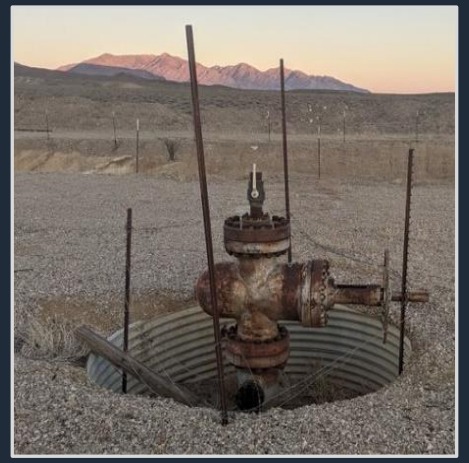


Roadmap

2021	2022-2023	2024	2025-2026	2026+
Proof of Concept & early testing	Thermal models, Lab build & demos, Shallow demos	Prototype well, brownfield development	First commercial projects, Scale development	Global deployment of utility scale projects

TRE slurry testing under pressure


1-meter lab demonstration



$13.5M Series A (2022)	$20M Series A Ext.	$65M Growth

$400M DevCo + AssetCo

Ready for scale and high temperatures

	Legacy geothermal	Enhanced geothermal (EGS)	Advanced geothermal (AGS, CLGS)	Thermal Reach Enhancement (TRE)
Concept	Multiple wells produce and re-inject fluid into natural, hot water reservoir (i.e., aquifers or hot springs)	Multiple wells produce and re-inject fluid into *man-made* hot water reservoir created through fracking	Closed-loop well(s); energy production based on volume of wellbore in contact with hot rock	Single closed-loop well; energy production based on interaction of conductive material with rock
Exploration risk	✗ High - 1 in 3 wells dry	~ Medium - need favorable geology	~ Medium - need favorable geology	✓ Low – all we need is hot rock
Water dependence	✗ High	✗ High	~ Medium – dependent on integrity of uncased well	✓ None – closed water loop contained by steel casing
Opex risk	✗ High - scale and corrosion drive re-work	✗ High – frac collapse	~ Medium - dependent on integrity of uncased well	✓ Low – engineered for zero subsurface opex
High-temperature readiness	~ Ready, but flow loss likely	✗ Not ready – tech needed (elastomers) non-operational >200°C	✗ Not ready – lateral drilling tech non-operational >200°C	✓ Ready – low risk well execution at high temp
Well construction risk	~ Medium - uncased well	~ Medium - reservoir stability	✗ High – complex well plan	~ Medium - proven drilling tech but new material placement

Clean.
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Forever.

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